Bingjie YAN

Medical Agents · Multimodal Ophthalmic AI · Federated Learning

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"Keep the curiosity."

EDUCATION

School of Optometry, The Hong Kong Polytechnic University — *Hong Kong, China*

Doctor of Philosophy — *2025.09 - Present*

- **Supervisor:** Prof. Danli SHI; **Co-supervisor:** Prof. Mingguang HE.
- **Research Topic:** Towards a Generalizable Multimodal Tool-Use Agent for Ophthalmology.

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS & UCAS) — *Beijing, China*

Master of Engineering, Electronic and Information Engineering — *2022.09 - 2025.06*

- **Supervisors:** Prof. Yiqiang CHEN and Prof. Xinlong JIANG. **GPA:** 3.79/4.
- **Research Topic:** Heterogeneous federated learning and foundation models for privacy-preserving healthcare.
- **Awards:** **National Scholarship (3%)**, Merit Student Awards (10%), The First Prize Scholarship (20%).

School of Computer Science and Technology, Hainan University (211) — *Hainan, China*

Bachelor of Engineering, Software Engineering for Big Data — *2018.09 - 2022.06*

- **GPA:** 3.68/4 (89.65/100). **Ranking:** 10/179.
- **Awards:** The First Prize Scholarship (3%), Outstanding Graduate Awards (3%) and Merit Student Awards.

RESEARCH INTERESTS

Medical Agents and Agentic AI: Clinical decision support and autonomous ophthalmic workflows.
Multimodal Ophthalmic AI: Vision-language benchmarks, report generation, and ocular world models.
Privacy-Preserving Federated Learning: Cross-silo training and model fusion.
Online Optimization and Learning Systems: Constrained optimization and asynchronous training.

RESEARCH EXPERIENCE

Institute of AI Industry Research, Tsinghua University

Research Intern — *2024.07 - 2024.10*

Hong Kong Baptist University

Research Assistant — *2024.06 - 2024.07*

FedML Inc.

Remote Research Intern — *2022.06 - 2022.09*

PUBLICATIONS

Full publication list: Homepage | Google Scholar. * Equal contribution; † corresponding author.

Selected Publications

[1] **FedEYE: A Scalable and Flexible End-to-end Federated Learning Platform for Ophthalmology**
Bingjie YAN*, Danmin CAO*, Xinlong JIANG, Yiqiang CHEN†, Weiwei DAI†, ..., Zhirui WANG
Patterns, Cell Press, JCR-Q1, IF=6.7, 2024 [Paper]

[2] **EyeAgent: An Agentic AI System for Multimodal Clinical Decision Support in Ophthalmology**
Danli SHI*, Xiaolan CHEN*, **Bingjie YAN***, ..., Mingguang HE†
NEJM AI, accepted; arXiv:2511.09394v3 (revised 2026) [arXiv]

[3] **Model Trip: Enhancing Privacy and Fairness in Model Fusion across Multi-Federations for Trustworthy Global Healthcare**
Qian CHEN, Yiqiang CHEN†, **Bingjie YAN**, ..., Qiang YANG
ICDE'24, CCF-A, 2024, Oral [Paper]

Medical Agents and Agentic AI

[4] **EyeFlow: A Self-Evolving Multi-Agent Framework for Knowledge-Grounded Ophthalmic Diagnosis**
Bingjie YAN, Xiaolan CHEN, Mingguang HE, Danli SHI
11th Asia Pacific Tele-Ophthalmology Society (APTOS) Symposium 2026, Free Paper Abstract [Abstract]

[5] **Empowering Locally Deployable Medical Agent via State Enhanced Logical Skills for FHIR-based Clinical Tasks**
Wanrong YANG, Zhengliang LIU, Yuan LI, **Bingjie YAN**, ..., Danli SHI
arXiv preprint, arXiv:2603.06902, 2026 [arXiv]

[6] **From visual question answering to intelligent AI agents in ophthalmology**
Xiaolan CHEN, Ruoyu CHEN, Pusheng XU, ..., **Bingjie YAN**, ..., Danli SHI
British Journal of Ophthalmology, 110(1), 1–7, 2026 [Paper]

Multimodal and Biomedical AI

[7] **EyeWorld: A Generative World Model of Ocular State and Dynamics**
Ziyu GAO, Xinyuan WU, Xiaolan CHEN, ..., **Bingjie YAN**, ..., Danli SHI
arXiv preprint, arXiv:2603.14039, 2026 [arXiv]
[8] **Benchmarking large multimodal models for ophthalmic visual question answering with OphthalWeChat**
Pusheng XU, Xia GONG, Xiaolan CHEN, ..., **Bingjie YAN**, ..., Danli SHI
Advances in Ophthalmology Practice and Research, 6(1), 33–41, 2026 [Paper]
[9] **EyeGraphGPT: Knowledge Graph Enhanced Multimodal Large Language Model for Ophthalmic Report Generation**
Zhirui WANG, Xinlong JIANG†, Chenlong GAO, ..., **Bingjie YAN**, ..., Yiqiang CHEN
IEEE BIBM'24, CCF-B, 2024, Oral [Paper]
[10] **Correlation-Driven Multi-Modality Graph Decomposition for Cross-Subject Emotion Recognition**
Wuliang HUANG, Yiqiang CHEN†, Xinlong JIANG, ..., **Bingjie YAN**, ..., Jianrong YANG
ACM MM'24, CCF-A, CORE-A, 2024, Poster* [Paper]
[11] **An Improved Method for the Fitting and Prediction of the Number of COVID-19 Confirmed Cases Based on LSTM**
Bingjie YAN, Jun WANG, Zhen ZHANG, Xiangyan TANG†, ..., Neal XIONG
Computers, Materials and Continua, SCI, 2020 [Paper]

Federated Learning and Privacy

[12] **FairFHTL: Achieving Task-Agnostic Fairness in Federated Hetero-Task Learning**
Teng ZHANG, Yiqiang CHEN, Xinlong JIANG, Wuliang HUANG, Qian CHEN, Chenlong GAO, Zhirui WANG, **Bingjie YAN**
IEEE International Conference on Multimedia and Expo (ICME 2025), pp. 1–6 [DOI]
[13] **Buffalo: Biomedical Vision-Language Understanding with Cross-Modal Prototype and Federated Foundation Model Collaboration**
Bingjie YAN, Qian CHEN, Yiqiang CHEN†, ..., Teng ZHANG
ACM CIKM'24, CCF-B, CORE-A (Acceptance Rate: 22.7%), 2024, Oral [Paper]
[14] **PrivFusion: Privacy-Preserving Model Fusion via Decentralized Federated Graph Matching**
Qian CHEN, Yiqiang CHEN†, Xinlong JIANG, ..., **Bingjie YAN**, ..., Bo YE
IEEE Transactions on Knowledge and Data Engineering (TKDE), CCF-A, CORE-A, 2024* [Paper]
[15] **FedTAM: Decentralized Federated Learning with a Feature Attention Based Multi-teacher Knowledge Distillation for Healthcare**
Tingting MOU, Xinlong JIANG†, Jie LI, **Bingjie YAN**, ..., Yiqiang CHEN
IEEE ICPADS, CCF-C, 2023, Oral [Paper]
[16] **FedCM: A Real-time Contribution Measurement Method for Participants in Federated Learning**
Bingjie YAN, Boyi LIU†, Lujia WANG, Yize ZHOU, Zhixuan LIANG, Ming LIU, Cheng-Zhong XU
IJCNN, CCF-C, 2021 [Paper]
[17] **Experiments of Federated Learning for COVID-19 Chest X-ray Images**
Bingjie YAN, Jun WANG, Jieren CHENG†, ..., Boyi LIU
ICAIS, EI, 2021 [Paper]

Foundation Models and Knowledge Distillation

[18] **Survey on Knowledge Distillation for Large Language Models: Methods, Evaluation, and Application**
Chuanpeng YANG, Yao ZHU, Wang LU, Yidong WANG, Qian CHEN, Chenlong GAO, **Bingjie YAN**, Yiqiang CHEN
ACM Transactions on Intelligent Systems and Technology, 16(6), Article 143, 2025 [Paper]
[19] **Towards Harnessing the Collaborative Power of Large and Small Models for Domain Tasks**
Yang LIU, **Bingjie YAN**, Tianyuan ZOU, ..., Ya-Qin ZHANG
arXiv preprint, arXiv:2504.17421, 2025 [arXiv]

Online Optimization and Computing Systems

[20] **Doubly-Bounded Queue for Constrained Online Learning: Keeping Pace with Dynamics of Both Loss and Constraint**
Juncheng WANG, **Bingjie YAN**, Yituo LIU
Proceedings of the AAAI Conference on Artificial Intelligence, 39(20), 21135–21143, 2025 [Paper]
[21] **Im2col-Winograd: An Efficient and Flexible Fused-Winograd Convolution for NHWC Format on GPUs**
Zhiyi ZHANG, Pengfei ZHANG, Zhuopin XU, **Bingjie YAN**, Qi WANG†
ICPP'24, CCF-B, 2024, Oral [Paper]

[22] **UbiMeta: A Ubiquitous Operating System Model for Metaverse**
Yiqiang CHEN†, Wuliang HUANG, Xinlong JIANG, ..., **Bingjie YAN**, ..., Guodong LONG
International Journal of Crowd Science (IJCS), 2023 [Paper]

[23] **AFL-CS: Asynchronous Federated Learning with Cosine Similarity-based Penalty Term and Aggregation**
Bingjie YAN, Xinlong JIANG†, Yiqiang CHEN†, Chenlong GAO, Xuequn LIU
IEEE ICPADS, CCF-C, 2023, Oral [Paper]

SELECTED AWARDS

2025 **Scholarship**, Institute of Computing Technology Ziqiang International Exchange Scholarship
2024 **National Scholarship (3%)**, Institute of Computing Technology, Chinese Academy of Sciences
2024 **First-Class Scholarship**, Institute of Computing Technology, Chinese Academy of Sciences
2022 **Outstanding Graduate**, Hainan University
2019 **First-Class Scholarship**, Hainan University
2017 **Silver**, (Intl.) 11th Asia and Pacific Informatics Olympiad, APIO — *Beijing*
2019 **1st Prize**, (Natl.) The 3rd Silk Road Robotics Innovations Competition — *Xi'an*
2020 **2nd Prize**, (Natl.) Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — *Beijing*
2020 **2nd Prize**, (Natl.) China Collegiate Computing Contest - Group Programming Ladder Tournament — *China*
2020 **2nd Prize**, (Natl.) Chinese Collegiate Computing Competition — *Beijing*
2023 **Silver**, (Natl.) China International College Students' "Internet+" Innovation and Entrepreneurship Competition — *Beijing*
2020 **Gold & Silver**, (Prov.) The 6th "Internet+" Innovation and Entrepreneurship Competition in Hainan — *Hainan*

PROFESSIONAL ACTIVITIES

Conference Reviewer

- Served as a reviewer for MICCAI (2026), AAAI (2025–2027), and ICME (2025).

Teaching Assistant

- *SO1D02: AI in Healthcare*, The Hong Kong Polytechnic University (Semester 1, 2026/27).

Leadership

- President, IEEE Hainan University Branch (Mar. 2021–Jun. 2022).
- Co-founder and Vice President, Association of Robotics and Artificial Intelligence (Jul. 2020–Jun. 2022).
- Vice President, Cyberspace Security Association at Hainan University (Sep. 2020–Jun. 2021).

SKILLS & INTERESTS

Programming	Python, C/C++, Java, JavaScript, TypeScript, HTML
ML & Training	PyTorch, TensorFlow, Hugging Face Transformers, PEFT (LoRA), timm, DeepSpeed
Agents & Serving	LangChain, LangGraph, MCP, RAG, vLLM, SGLang, Ollama
Tools & Data	FastAPI, Gradio, React, Git, Docker, Linux, NumPy, pandas
Language	Chinese (Native), English (Fluent), IELTS: 6.5 (L:8.5, R:6.5, W:6.0, S:5.5), CET-6: 478, CET-4: 539
Photography	Enjoy the life and capture the moments ;)